# Bond Market Response to FOMC Communication: Overreaction and Diagnostic Expectations

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**Abstract**

Since 2004, the Federal Reserve has released the minutes of Federal Open Market Committee (FOMC) meetings three weeks after each policy announcement, providing a natural setting to study how financial markets process sequential policy signals. This paper examines how the yield curve responds to both FOMC announcements and the subsequent release of meeting minutes. Using high-frequency identification, I extract monetary policy shocks and decompose them into target rate, forward guidance, and yield-curve "twist" components. I document a systematic overreaction of yields to the forward guidance component of announcements, followed by a partial reversal when the corresponding minutes are released. I interpret this pattern through a model of diagnostic expectations, in which investors initially overweight salient forward guidance signals and later revise their beliefs as additional context becomes available in the minutes.

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# 1 Introduction

Beginning in December 2004, the Federal Open Market Committee (FOMC) began releasing the minutes of its meetings approximately three weeks after each policy decision. These minutes have since become a key source of monetary policy information, offering detailed insights into the Committee's economic outlook, risk assessments, and policy deliberations. They allow market participants to refine their understanding of the policy stance and form expectations about future actions. For instance, on May 25, 2022, CNBC reported that "Fed minutes point to more rate hikes that go further than the market anticipates," referring to the minutes of the May 4 meeting released earlier that day.

While an extensive literature examines the financial market response to FOMC announcements and statements, the information content of the minutes has received far less attention. The staggered release of information by the FOMC provides a natural setting to study how financial markets process FOMC communication. Since FOMC minutes carry contextual information about the Fed's policy deliberations, the information content is qualitatively different from policy announcements. Do financial market participants' interest-rate expectations respond in a qualitatively different way to these two modes of communication?

This paper aims to shed light on what I call the "FOMC minutes effect": the incremental influence of minutes on the yield curve over and beyond the FOMC announcement. Using high-frequency changes in medium-to-long term U.S Treasury yields around monetary policy events, I identify monetary policy shocks associated with both types of Fed communication. I find a statistically significant negative relationship between the response of medium-term treasury yields (2- to 5-year maturities) to FOMC announcements and to the corresponding FOMC minutes. Since FOMC announcements are publicly available three weeks prior to the release of FOMC minutes, this finding provides evidence against full-information rational expectations. Using a factor model, I decompose the shifts in the yield curve in response to FOMC communication into two factors that can be interpreted as a forward guidance and a yield curve twist factor. The evidence suggests that bond market participants overreact to forward guidance in announcements and subsequently correct after the release of minutes. I interpret this pattern using a model of noisy information and diagnostic expectations, where agents initially overweigh salient signals in the form of FOMC announcements and later revise beliefs as more detailed policy information becomes available upon the release of the minutes. I do not find a statistically significant relationship between announcement and minutes-day shifts in the "twist" factor.

## 2 Related literature

The present work borrows insights from the literature on high-frequency identification of monetary policy shocks pioneered by Kuttner (2000) and more closely to work by Gurkaynak, Sack and Swanson (2004) as well as Swanson (2021). These papers establish high-frequency shifts in asset prices as important tools for the identification of monetary policy shocks. Broadly, these studies use shifts in Fed funds futures in a 30-minute window around monetary policy announcements to measure the underlying shock or surprise. Critical to this method is the idea that the response of asset prices within a short event-window of monetary news or communication would be uncorrelated with other shocks, and hence allows for clean identification of monetary policy shocks I use a variant of Swanson (2021)'s method for identification of the components of monetary policy shocks in this paper.

Now I turn to literature specifically looking at the effects of FOMC minutes on asset prices. Jegadeesh and Wu (2015) find evidence that the language of the minutes contain information above and beyond the announcement that is relevant to financial markets. Nechio and Wilson (2016) also make use of text analysis to find that impact of minute releases on asset prices are largest when the "tone" of the minutes differs from that of the announcement, thus relaying more additional information. Among the high-frequency identification literature Rosa (2013) studies the volatility of asset-prices around a 30-minute window around the release of FOMC minutes and finds them to be significantly different from non-event days. However, while these studies show that minutes-day effects are significant, they do not study what kind of information financial market participants actually glean from them.

This paper also invokes ideas from a strand of literature studying forecast revisions and forecaster irrationality. Beginning with Coibion and Gorodnichenko (2012), several papers (Kohlhas and Walther (2021), Bordalo, Gennaioli, Ma and Shleifer (2020), Angeletos, Huo and Sastry (2021)) address the problem of irrational behavior in forecast revisions on receiving macroeconomic news. Coibion and Gorodnichenko (2012) regress professional forecasters' forecast revisions upon receiving information about macroeconomic variables on the realized forecast errors. Their findings of autocorrelated forecast errors and a positive correlation between forecast revisions and realized errors point towards information rigidities, in violation of Full Information Rational Expectations (FIRE). They find that for inflation and several other macro variables, consensus forecasts under-react relative to FIRE. Bordalo, Gennaioli, Ma and Shleifer (2020) (henceforth - BGMS) find evidence for consensus-level overreaction and individual-level under-reaction to news among forecasters. In this paper, I model the response of bond market participants to FOMC announcements and minutes in the same spirit and address the broad patterns observed.

# 3 Measuring Monetary Policy Shocks

I use daily data on U.S treasury yields[1] over the sample period beginning in 2008 and ending in 2024.[2] Around a monetary policy event, information is conveyed to private-sector agents that causes them to revise their assessment of the stance of policy. In this paper, shocks are measured based on shifts in the yield curve in a 1-day window[3] around the event. I use a longer window than the traditional 30-minutes to allow for the market to completely interpret and respond to the information contained in the announcements as well as minutes. Rosa (2013) documents that the absorption of news is more prolonged, taking roughly one hour and thirty minutes, compared with the time associated with the release of the FOMC minutes. Gurkaynak et al. (2004) show that a one-day window is sufficient for the identification of monetary policy shocks after 1995. My approach is similar to Inoue and Rossi (2021) who use daily changes in yields to derive a functional measure of monetary policy shocks, defined as the shift in the entire yield curve around a monetary policy announcement.

Let the shift in yields at maturity $\tau$ around a monetary policy event be given by $\Delta y(\tau)$:

$$\Delta y(\tau) = \Delta y^{mp}(\tau) + \varepsilon(\tau) \tag{1}$$

where $\Delta y^{mp}(\tau)$ denotes the systematic response to monetary policy communication. It is confounded by noise or measurement error $\varepsilon(\tau)$, arising from stray movements in the yield curve that are unrelated to the monetary policy event. The ratio of variance of the systematic component $\Delta y^{mp}(\tau)$ to that of the noise $\varepsilon(\tau)$ represents the signal-to-noise (SNR) ratio in the measurement of the monetary policy shock.

## 3.1 Announcement- and minutes-day shifts in the yield curve

The sample period includes 134 FOMC announcements and corresponding minutes. Table 1 summarizes the standard deviations and the mean absolute changes in yields on days corresponding to FOMC announcements and the release of FOMC minutes.

We see that at the shorter maturities, minutes-releases do not move treasury yields much as compared to FOMC announcements. At longer maturities, the effects are comparable. This is

[1] the data on constant maturity par yields is obtained from the U.S Department of Treasury website

[2] The Fed began the release of its Survey of Economic Projections after October 2007. Since the inclusion of these projections may affect how Fed communication is interpreted by financial markets, I restrict the sample period to the years 2008-2024. I repeat the analysis for alternative choices of sample periods: keeping the whole sample period from 2005 to 2024 as well as beginning the sample after 2011, since this is the year that the Fed chair began holding a press conference immediately following FOMC announcements. I find that the results remain qualitatively similar across these exercises.

[3] closing 1-day after the event minus closing 1-day before

Table 1: Summary of changes in yields around FOMC announcements and minutes

| | Announcement day | | Minutes day | |
|---|---|---|---|---|
| Maturity | Std. Dev. | Mean Abs. | Std. Dev. | Mean Abs. |
| 1m | 0.146 | 0.049 | 0.072 | 0.032 |
| 3m | 0.108 | 0.040 | 0.035 | 0.020 |
| 6m | 0.064 | 0.033 | 0.036 | 0.019 |
| 1y | 0.060 | 0.037 | 0.048 | 0.028 |
| 2y | 0.088 | 0.061 | 0.072 | 0.048 |
| 5y | 0.108 | 0.085 | 0.089 | 0.066 |
| 7y | 0.115 | 0.089 | 0.090 | 0.068 |
| 10y | 0.110 | 0.086 | 0.089 | 0.067 |

There are 136 FOMC announcements and minutes-releases in the sample period considered. All values are expressed in basis points.

intuitive if we consider the announcements to be informative of changes in the Federal Reserve's near-term policy stance. FOMC minutes may carry information relevant only to medium-to-long term behavior of interest rates. Hence, in the rest of the paper, I restrict focus to medium-to-long maturities (1-year and longer).

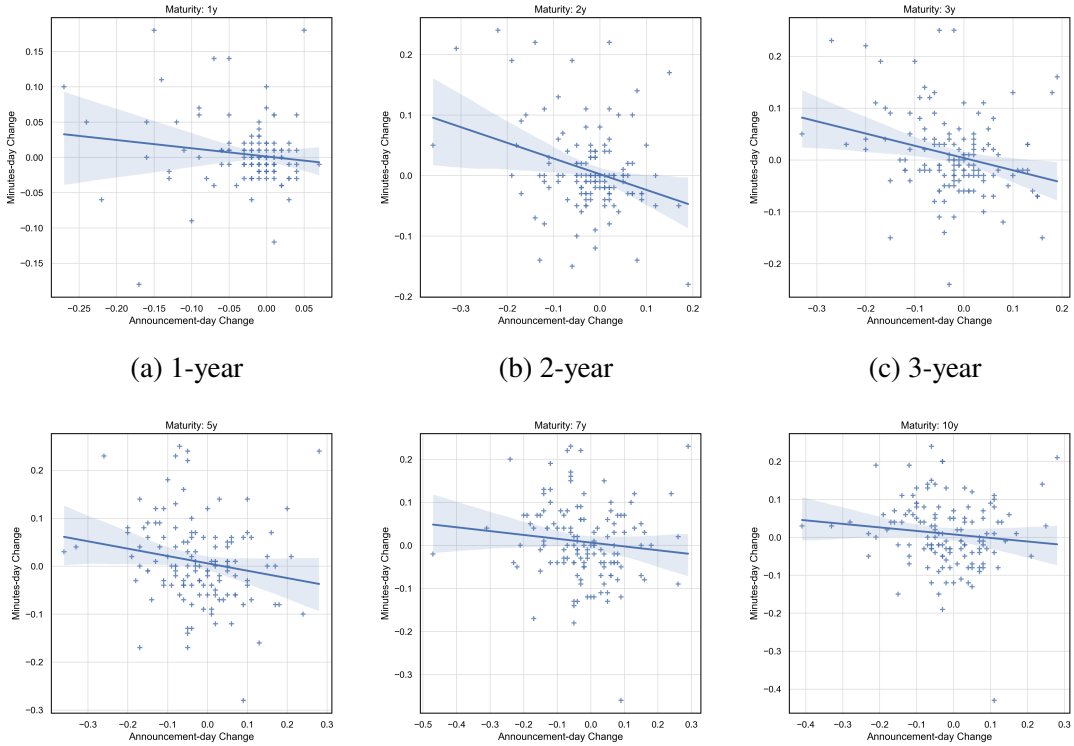
For each FOMC meeting indexed by $t$, we observe the changes in yields at several maturities corresponding to the announcement as well as the minutes. For each meeting, I denote the change in the yield at maturity $\tau$ in response to the announcement and the minutes as $\Delta y_t^{Ann}(\tau)$ and $\Delta y_t^{Min}(\tau)$ respectively. Hence, for each meeting $t$, we obtain vectors $\Delta Y_t^{Ann} = [\Delta y_t^{Ann}(\tau_1), \Delta y_t^{Ann}(\tau_2), \ldots, \Delta y_t^{Ann}(\tau_k)]$ and $\Delta Y_t^{Min} = [\Delta y_t^{Min}(\tau_1), \Delta y_t^{Min}(\tau_2), \ldots, \Delta y_t^{Min}(\tau_k)]$ capturing the announcement and minutes shocks.

# 4 Correlation between announcement-day and minutes-day responses

Given that the FOMC makes monetary policy announcements prior to releasing the minutes of the FOMC meetings, at the time that the minutes are released, the corresponding announcement is already publicly known. Under full-information rational expectations (FIRE), the shift in yields at any maturity upon the release of FOMC minutes must be uncorrelated with those around an FOMC announcement. Thus, under FIRE, we would expect the slope of the regression between announcement-day and minutes-day changes in yields to be zero.

Figure **??** plots the announcement-day and minutes-day responses of yields at various maturities. Visually, we see that they are negatively correlated. Table 2 describes the results of re-

gressing minutes-day shifts against corresponding announcement-day shifts in yields at various maturities for the sample period.




(a) 1-year (b) 2-year (c) 3-year

(d) 5-year (e) 7-year (f) 10-year

Figure 1: Comparing minutes-day and announcement-day changes in yields at various maturities

Table 2: Coefficient of the regression:
$\Delta y_t^{Min}(\tau) = \alpha + \beta \Delta y_t^{Ann}(\tau) + \epsilon_t(\tau)$

| Maturity ($\tau$) | Coefficient ($\beta$) | Std. Err. | $R^2$ | $N$ |
|---|---|---|---|---|
| 1y | -0.117 | (0.070) | 0.021 | 134 |
| 2y | -0.259** | (0.071) | 0.091 | 134 |
| 3y | -0.236** | (0.073) | 0.073 | 134 |
| 5y | -0.153** | (0.071) | 0.034 | 134 |
| 7y | –0.089 | (0.069) | 0.012 | 134 |
| 10y | -0.093 | (0.071) | 0.013 | 134 |

Each row reports the coefficient from a separate regression of minutes-day yield shifts on announcement-day yield shifts for the indicated maturity. Standard errors are shown in parentheses. ** $p < 0.05$.

# 5 A factor model

While shifts in the yields at various maturities can capture monetary policy surprises, they are known to be highly correlated. Instead, they are well-summarized by a small number of common *factors* or underlying drivers. Using a factor model to decompose the vector of changes in yields can offer a low-dimensional representation of the informational contained in the vector of shifts in the yield curve across maturities, while offering a sparse set of econonmically interpretable signals.

Identification of the shocks borrows from Swanson (2021). Shifts in the yield curve at $n$ maturities in a one-day window around monetary policy event (FOMC announcement or minutes release) are collected into a $T \times n$ matrix $X$.[4] Hence each column represents the change in yields at the 1-year, 2-year, 3-year, 5-year, 7-year and 10-year maturities. Each row corresponds to a monetary policy event, i.e, an FOMC announcement or a FOMC minutes release. The matrix $X$ is first standardized so that all columns have zero mean and unit variance.

Principal Component Analysis (PCA) involves factoring a $T \times n$ matrix $X$ as:

$$X = F\Lambda + \varepsilon \tag{2}$$

where $F$ is a $T \times k$ matrix of $k \leq n$ factors. $\Lambda$ is a $k \times n$ the matrix of the loadings for the different maturities on the factors.

To determine the appropriate number of factors $k$, I conduct the Cragg-Donald test (see Cragg and Donald (1997)), following Swanson (2021) and Gurkaynak et al. (2004). Given a null hypothesis of rank $k = k_0$ versus an alternative $k > k_0$, the Cragg-Donald test searches over all possible factor models with $k_0$ factors to find the one that brings the residuals $\varepsilon$ as close to uncorrelated white noise as possible; the test then measures the distance between the residuals and white noise using a Wald statistic. There are 134 FOMC announcements and corresponding minutes in the sample period from January 2008 to December 2024, totaling to 268 monetary policy events. The data includes shifts in the yield curve across 6 maturities, yielding a matrix $X$ of dimensions $268 \times 6$. The Cragg-Donald test suggests two factors to be important.

I find that the first two principal components capture 98% of the variation in $X$, hence providing a good fit to the data. However, this factorization is not unique. We can see this by considering any rotation with orthogonal matrix $U$ yielding $\tilde{F} = FU$ and $\tilde{\Lambda} = U'\Lambda$, which explain the shifts in yields equally well. All such models are observationally equivalent. To aid identification, we must impose structural restrictions so that the factors are interpretable as the forward guidance and the yield curve twist factors. I impose restrictions on the factors' effects on the changes in

[4] The window begins at the market-close of the preceding day ends at market-close the following working day.

yields at the various maturities based on economic intuition so that they can be interpreted as the *forward guidance* and the *yield curve twist* factor:

- the forward guidance factor has a positive effect on yields at maturities 1y to 10y
- the twist factor has a positive loading on maturities up to a threshold and negative loading on longer maturities; this restriction allows us to interpret this factor as having a twist effect on the yield-curve

The restrictions are intuitive: since forward guidance is defined as the component of policy that is relevant to the future path of policy, so it moves yields at all maturities in the same direction. The second factor has the effect of lowering the long-term yields but raising the yields at the shorter-term maturities, thereby twisting the yield curve. The Fed has engaged in such operations where short-to-medium term treasuries are sold and long-term treasuries are purchased without expanding the Fed's balance sheet.

I then search over all orthogonal rotations $U$ and choose the angle that makes the first column of $U'\Lambda$ have the same sign and the second column to have exactly one sign switch.

Table 3: Factor Loadings

| Maturity | Forward Guidance | Twist |
|---|---|---|
| 1y | 0.2017 | 0.4687 |
| 2y | 0.3486 | 0.5455 |
| 3y | 0.4111 | 0.3377 |
| 5y | 0.4751 | -0.0661 |
| 7y | 0.4864 | -0.3405 |
| 10y | 0.4544 | -0.4985 |

*Notes:* Entries are factor loadings for different maturities. All values reported to four decimal places.

# 6 What information do minutes convey?

In Figure 2, I plot the forward guidance component of the announcement shock against the corresponding minutes shock for each FOMC meeting in the sample. **??** reports the result of the regression.

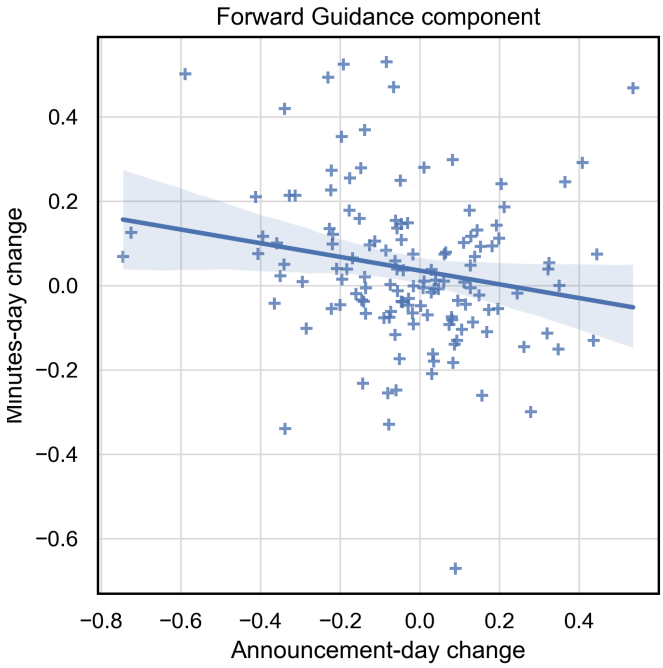



Figure 2: Forward guidance factor: Negative relationship between announcement- and minutes-day responses

Table 4: Results of the regression:
$\Delta fg_t^{min} = \alpha + \beta \Delta fg_t^{ann} + \epsilon_t$

| | Coefficient | Std. Err. | p-value |
|---|---|---|---|
| $\alpha$ | $0.0350^{**}$ | 0.013 | 0.005 |
| $\beta$ | $-0.1625^{**}$ | 0.074 | 0.029 |
| Observations | | 134 | |
| R-squared | | 0.037 | |
| Adj. R-squared | | 0.030 | |
| F-statistic | | 4.789 | |
| Durbin–Watson | | 2.136 | |

Notes: Robust standard errors (HAC, 4 lags) in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The results are indicative of a statistically significant, negative relationship between the forward guidance component of the announcement and minutes shocks. This prompts a rejection of Full Information Rational Expectations (FIRE) since under FIRE, shifts in the yield curve should be

uncorrelated with all past information. Rather, the results show that agents systematically *overreact* when interpreting the forward guidance in the FOMC announcements and backpedal once the minutes are released. Looking at the effects on individual maturities (shown in Appendix A), a similar pattern is revealed, with medium-term yields showing a pattern of overreaction.

For the twist component, the minutes and announcement day effects are uncorrelated (see Figure 3 and Table 5). One plausible interpretation of this fact could be that since the quantum of Fed asset purchases and portfolio re-balancing is revealed during the announcement itself, minutes do not carry much additional information relevant to long-term yields (assuming that shifts in liquidity and term premia are small). However, the medium-term yields would be impacted by information relevant to policy deliberations over the coming FOMC meetings. Hence the information contained in FOMC minutes could be interpreted by bond market participants as being relevant to incipient changes in policy over the next few quarters.

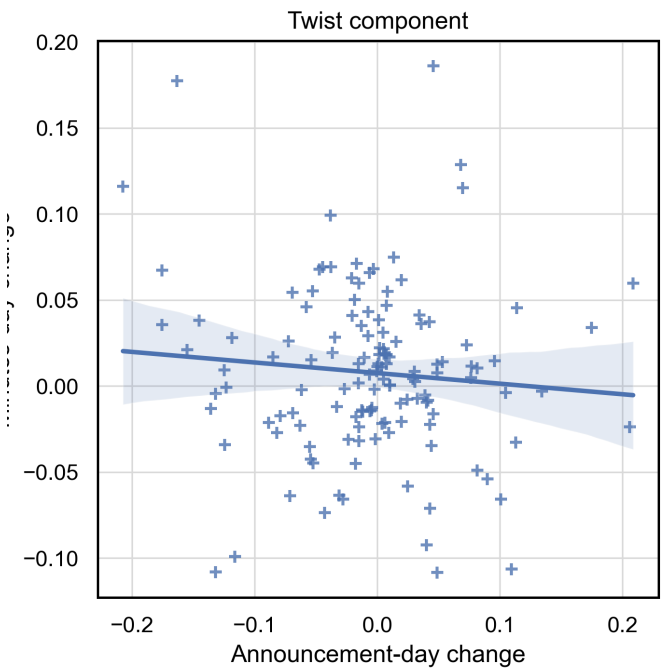



Figure 3: Twist factor: No relationship between announcement- and minutes-day responses

Table 5: Results of the regression:
$\Delta twist_t^{min} = \alpha + \beta \Delta twist_t^{ann} + \epsilon_t$

| | Coefficient | Std. Err. | p-value |
|---|---|---|---|
| $\alpha$ | 0.0075 | 0.004 | 0.072 |
| $\beta$ | $-0.0615$ | 0.068 | 0.366 |
| Observations | | 134 | |
| R-squared | | 0.008 | |
| Adj. R-squared | | 0.001 | |
| F-statistic | | 0.817 | |
| Durbin–Watson | | 2.162 | |

Notes: Robust standard errors (HAC, 4 lags) in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

# 7 Model

In this section, I present a model of belief updates under diagnostic expectations following Bordalo et al. (2020). Denote the true period-$t$ FOMC forward guidance stance as the fundamental $x_t$, following an exogenous process. Each period, the FOMC generates two signals carrying information about $x_t$: the announcement signal, denoted by $s_t^a$ followed by a minutes signal $s_t^m$.

A Bayesian, or rational, forecaster enters period $t$ carrying from the previous period beliefs about $x_t$ summarized by a probability density $f(x_t|S_{t-1})$ where $S_{t-1}$ denotes the full history of signals observed by this forecaster. In period $t$ , the forecaster observes a new signal $s_t$ in light of which he updates his estimate of the current state using Bayes' rule:

$$f(x_t|S_t) = \frac{f(s_t|x_t)f(x_t|S_{t-1})}{\int f(s_t|x)f(x|S_{t-1})dx}$$

## 7.1 A simple model of noisy signals

At the beginning of period $t$, the FOMC makes an announcement, generating a noisy signal $s_t^a$:

$$s_t^a = x_t + e_t^a, \quad e_t \sim N(0, \sigma_a^2)$$

Here, $x_t$ denotes the forward guidance stance of the Fed's policy at $t$. Upon observing $s_t^a$, the rational forecaster revises her beliefs regarding the underlying state $x_t$ using Bayes' rule, arriving

at the estimate $\mathbb{E}[x_t|S^{t-1} \cup s_t^a] = x_{t|t}$.

I assume that the uncertainty about the fundamental $x_t$ is completely resolved when minutes are released and market participants observe the true underlying forward guidance stance $x_t$. Hence, once the minutes signal denoted by $s_t^m$ is observed, the true $x_t$ is effectively revealed:

$$s_t^m = x_t$$

Thus, the release of FOMC minutes leads to the realization of the forecast error $RE_t = x_t - x_{t|t}$. This denotes the extent of mistakes in their initial assessment based on the announcement alone.

Assume that $x_t$ follows a persistent AR(1) process given by:

$$x_t = \rho x_{t-1} + u_t, \quad u_t \sim N(0, \sigma_u^2)$$

Hence, upon observing all signals upto and including period $t$, the rational forecaster's expectations satisfy: $\mathbb{E}[x_t|S_t] = x_t$ and $\mathbb{E}[x_{t+1}|S_t] = \rho x_t$,

The shift in the forward guidance factor upon period-$t$ FOMC announcement is hence given by:

$$\Delta fg_t^{ann} = \mathbb{E}[x_t|S^{t-1} \cup s_t^a] - \mathbb{E}[x_t|S^{t-1}] = x_{t|t} - \rho x_{t-1}$$

Since upon observing the minutes, all uncertainty is resolved, the minutes-shock is simply:

$$\Delta fg_t^{min} = x_t - \mathbb{E}[x_t|S^{t-1} \cup s_t^a] = x_t - x_{t|t}$$

## 7.2 Rational Benchmark

A rational agent updates beliefs about the underlying state $x_t$ according to Bayes' Rule as:

$$f(x_t|S_t) = \frac{f(s_t|x_t)f(x_t|S_{t-1})}{\int_X f(s_t|x_t)f(x_t|S_{t-1})}$$

The rational estimate in this case is given by

$$f(x_t|S_t) \sim N(x_{t|t}, \frac{\Sigma\sigma_a^2}{\Sigma + \sigma_a^2})$$

where $\Sigma$ is the asymptotic variance of the prior $f(x_t|S_{t-1})$, $\Sigma_{t+1|t} = \Sigma_{t|t-1} = \Sigma$. In this case, the steady-state variance of $f(x_t|S_{t-1})$ is just $\sigma_u^2$.

$x_{t|t}$ evolves according to the Kalman filter as:

$$x_{t|t} = x_{t|t-1} + \frac{\sigma_u^2}{\sigma_u^2 + \sigma_a^2}(s_t - x_{t|t-1}) \tag{3}$$

Under Bayesian learning, the forecaster's time-$t$ estimate $x_{t|t}$ equals her prior $x_{t|t-1}$ plus a "gain" times the observed discrepancy $(s_t - x_{t|t-1})$. Further, $x_{t|t-1} = \rho x_{t-1}$

Define the Kalman gain, $K = \frac{\Sigma}{\Sigma+\sigma_\epsilon^2}$ where $0 < K < 1$

### 7.2.1 Diagnostic Distortion from Rationality

Following BGMS, I describe a model where bond market participants update their beliefs in a way consistent with the diagnostic heuristic. The diagnostic expectations model is based on the heuristic that after observing a signal that is different from their ex-ante expectation, states that were more likely to have generated the signal (representative) are given more weight than under Bayesian rationality.

Define

$$R(x) = \frac{f(x|S_{t-1} \cup s_t)}{f(x|S_{t-1})}$$

Under diagnostic expectations, the posterior distribution is given by

$$f^\theta(x|S_t) = f(x|S_t)R(x)^\theta \frac{1}{Z}$$

where $Z$ is a normalizing constant chosen so that $f^\theta$ integrates to 1 over the support of $x$. $\theta = 0$ corresponds to rational expectations, where the agent's beliefs correspond with the Bayesian probability distribution $f(x_t|S_t)$. For $\theta > 0$, the diagnostic filter inflates the probability of representative states (the ones were more likely to have generated the signal) and deflates the probability of the unrepresentative ones. Diagnostic beliefs are given by a normal distribution with a shifted mean:

$$f^\theta(x_t|S_t) \sim N(x_{t|t}^\theta, \frac{\sigma_u^2\sigma_\epsilon^2}{\sigma_u^2 + \sigma_\epsilon^2})$$

where the diagnostic state estimate $x_{t|t}^\theta$ is given by

$$x_{t|t}^\theta = x_{t|t-1} + (1+\theta)\frac{\Sigma}{\Sigma + \sigma_\epsilon^2}(s_t^a - x_{t|t-1}) \tag{4}$$

Under diagnostic expectations, the belief revision at $t$, $\Delta fg_t^{ann}$ is given by:

$$x_{t|t}^\theta - x_{t|t-1}^\theta = (1+\theta)(x_{t|t} - x_{t|t-1}) - \theta\rho(x_{t-1|t-1} - x_{t-1|t-2}) \tag{5}$$

The update $\Delta fg_t^{min}$ upon receiving the period-$t$ minutes signal which reveals $x_t$, is denoted by $RE_t$ and given by:

$$x_t - x_{t|t}^{\theta} = (\frac{1-K}{K} - \theta)(x_{t|t} - x_{t|t-1})$$

————-INCOMPLETE————————

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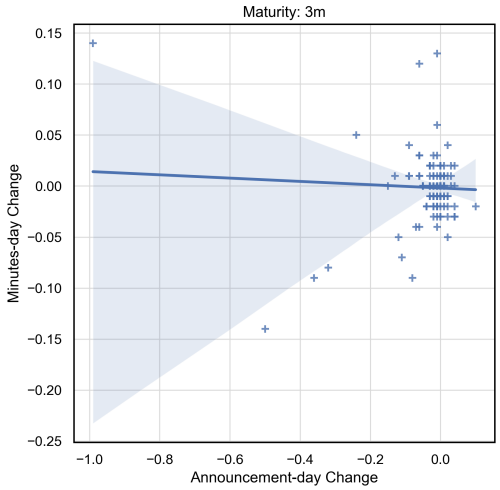
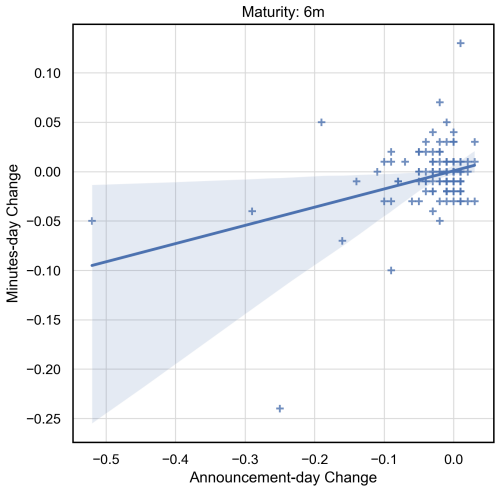
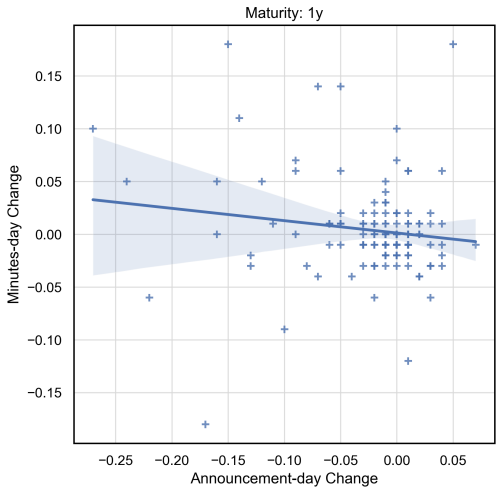
# A FOMC minutes and announcement effects at different maturities



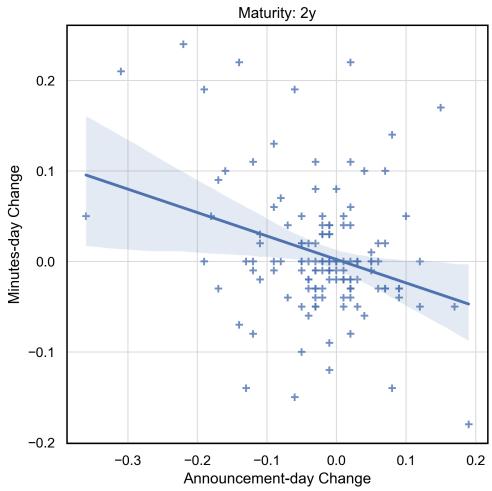

(a) 3-month




(b) 6-month

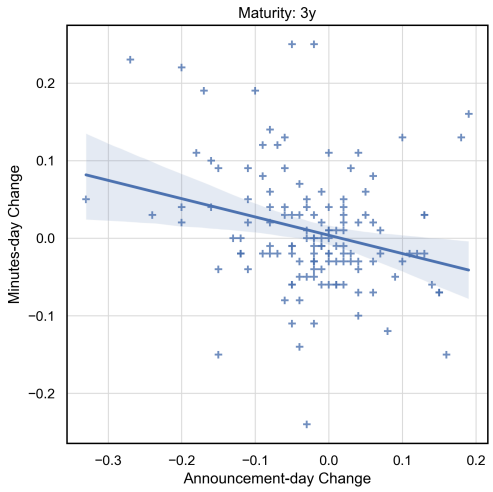



(c) 1-year

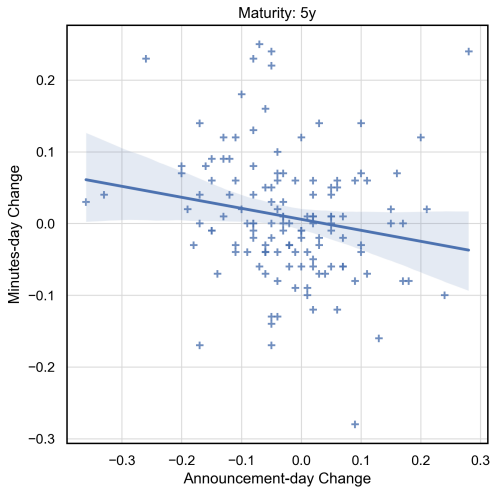



(d) 2-year

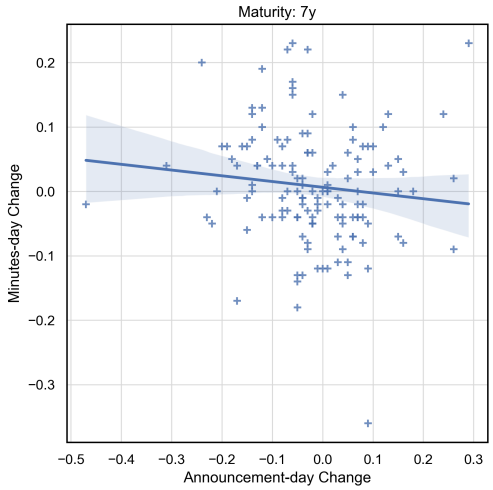
Figure 4: Comparing minutes-day and announcement-day effects at various maturities



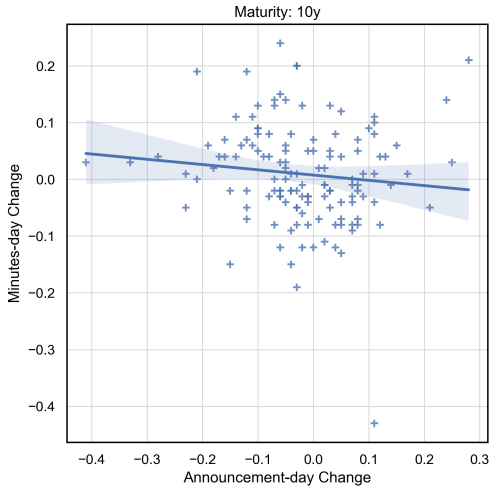

(a) 3-year




(b) 5-year




(c) 7-year




(d) 10-year

Figure 5: Comparing minutes-day and announcement-day shifts at various maturities

# B Does FOMC communication cause abnormal volatility in the yield curve?

## B.1 Do FOMC minutes move the yield curve?

If shifts in the yield curve around days corresponding to FOMC communication carry substantial information relevant to bond markets, one would expect the average magnitude of shifts in the yield curve on these days to be larger than those around regular days (not corresponding to any monetary news). One way to measure the magnitude of shifts in the yield curve is to measure the variance of changes in U.S treasury yields across maturities in a 1-day window around days corresponding to monetary policy events, such as FOMC announcements or release of minutes, and compare them against a "control" group of days without either of these events. To test this, a one-tailed F-test of variances is conducted for shifts in yields across maturities (ranging from 1-month to 10-years) testing the alternate hypothesis of larger variance of changes around monetary policy events (FOMC announcements or minutes-releases) as compared to non-event days (reported in subsection B.1. I find that during FOMC announcement days, shifts in yields across all but one maturity (1-year) have larger variance, and the null of equality of variance is rejected.

On the days corresponding to the release of FOMC minutes, the null of equal variance is not rejected for yields at any maturity.

Now I turn to the question of whether the minutes contain additional information over and above the announcement that is relevant to the financial markets. Since the minutes signal corresponds to the same FOMC meeting, it must contain similar information as the FOMC announcement. Under Full-information Rational Expectations, the "minutes shock" as measured by changes in the yield curve around the release of FOMC minutes would not be predictable based on any prior publicly available information. We can test this implication of FIRE by testing for a relationship between the components of announcement shocks and the corresponding minutes shocks.

Table 6: One-tailed F-tests: Minutes-day movements in the yield curve vs Non-monetary news days

| Maturity | $Var_{Min}$ | $Var_{NN}$ | $F$ | p-value |
|---|---|---|---|---|
| 1m | 0.0088 | 0.0096 | 0.921 | 0.7496 |
| 3m | 0.0047 | 0.0045 | 1.034 | 0.3706 |
| 6m | 0.0027 | 0.0029 | 0.944 | 0.6784 |
| 1y | 0.0033 | 0.0032 | 1.036 | 0.3651 |
| 2y | 0.0057 | 0.0050 | 1.142 | 0.1098 |
| 3y | 0.0066 | 0.0057 | 1.154 | 0.0926* |
| 5y | 0.0077 | 0.0067 | 1.155 | 0.0922* |
| 7y | 0.0077 | 0.0069 | 1.117 | 0.1528 |
| 10y | 0.0074 | 0.0063 | 1.169 | 0.0748* |

*Notes:* Sample sizes are $n_1 = 160$ and $n_2 = 4958$ for all maturities. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The one-tailed F-test does not reject the null that the variance of yields on days corresponding to the release of FOMC minutes is the same as those on days without any FOMC news.

Table 7: One-tailed F-tests: Variance of shifts in the yield curve on announcement days vs days with no monetary policy related news (control)

| Maturity | $Var_{Ann}$ | $Var_{NM}$ | $F$ | p-value |
|---|---|---|---|---|
| 1m | 0.0196 | 0.0096 | 2.047 | 0.0000** |
| 3m | 0.0106 | 0.0045 | 2.344 | 0.0000** |
| 6m | 0.0040 | 0.0029 | 1.409 | 0.0007** |
| 1y | 0.0036 | 0.0032 | 1.130 | 0.1297 |
| 2y | 0.0075 | 0.0050 | 1.512 | 0.0000** |
| 3y | 0.0088 | 0.0057 | 1.531 | 0.0000** |
| 5y | 0.0109 | 0.0067 | 1.641 | 0.0000** |
| 7y | 0.0120 | 0.0069 | 1.753 | 0.0000** |
| 10y | 0.0110 | 0.0063 | 1.741 | 0.0000** |

*Notes:* Sample sizes are $n_1 = 160$ and $n_2 = 4958$ for all maturities. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The one-tailed F-test rejects the null that the variance of yields on days corresponding to FOMC announcements is the same as those on days without any FOMC news, except for the 1-year yield.